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3. SPRING CONSTANT OF A SPRING

AIM

To determine the spring constant of a spring by using the method of vertical oscillations

APPARATUS REQUIRED

Spring, rigid support, hook, 50 g mass hanger, 50 g slotted masses, stop clock, metre scale, pointer

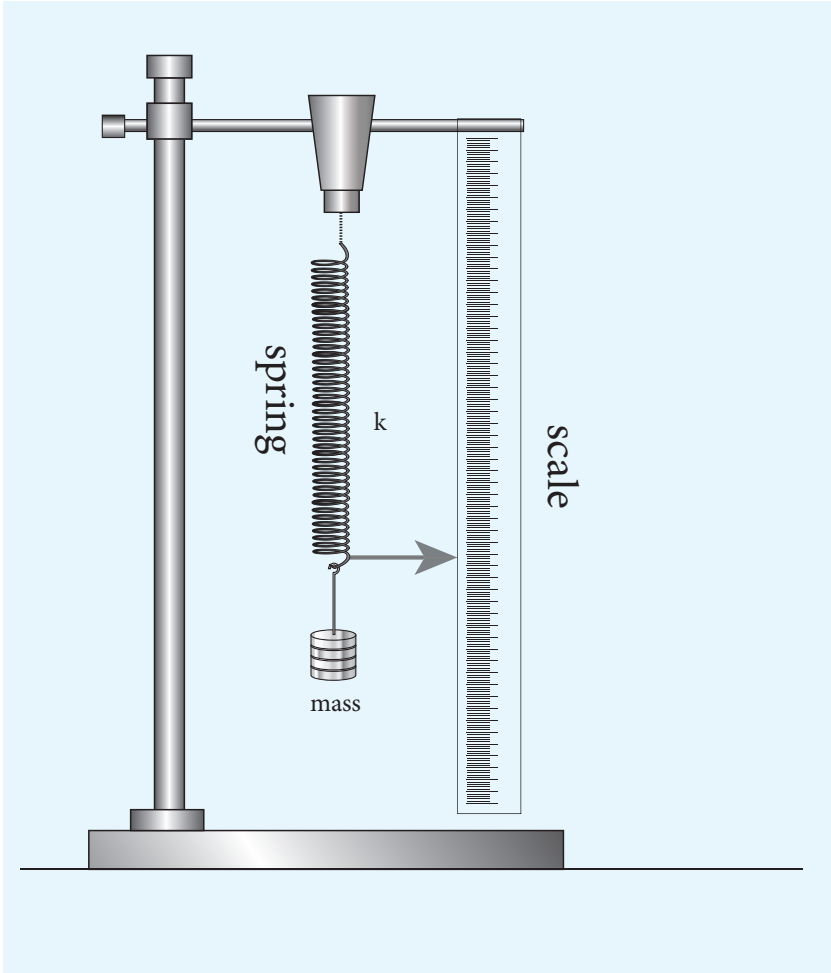
FORMULA

Spring constant of the spring $k = 4\pi^2 \left(\frac{M_2 - M_1}{T_2^2 - T_1^2} \right)$

where $M_1, M_2 \rightarrow$ selected loads in kg

$T_1, T_2 \rightarrow$ time period corresponding to masses M_1 and M_2 respectively in second

DIAGRAM



PROCEDURE

- A spring is firmly suspended vertically from a rigid clamp of a wooden stand at its upper end with a mass hanger attached to its lower end. A pointer fixed at the lower end of the spring moves over a vertical scale fixed.